

From Audio Description to Movement: Challenges and Design Principles in Video-based Body Exercise for Blind and Low Vision Users

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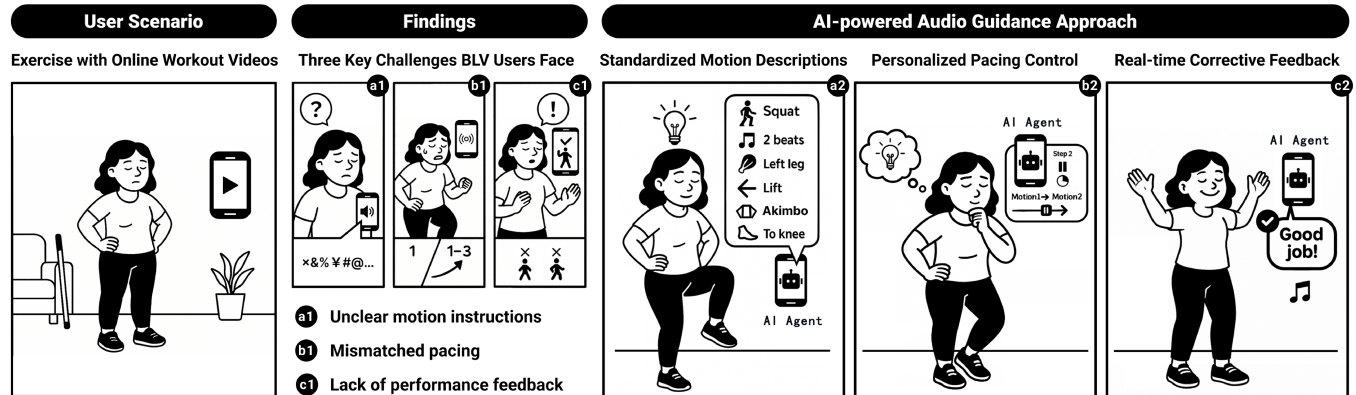


Figure 1: Our research reveals three key challenges blind and low-vision (BLV) users face when exercising with online workout videos: (a1) unclear motion instructions, (b1) mismatched pacing, and (c1) lack of performance feedback. To address these challenges, we propose an AI-powered audio-guided approach that improves workout video accessibility by offering (a2) standardized motion descriptions, (b2) personalized pacing control, and (c2) real-time corrective feedback—enabling BLV users to engage with online workout videos independently and effectively.

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Abstract

The proliferation of online workout videos has revolutionized fitness practices, yet their visual-centric design creates significant accessibility barriers for blind and low vision (BLV) users. Through interviews and co-watching sessions with eight BLV participants, we identify three key challenges in video-based exercise: (1) unclear motion instructions, (2) misaligned pacing, and (3) lack of performance feedback. To address these challenges, we propose five design principles for accessible video-based exercise: clear movement

descriptions, step-by-step breakdown, rhythm guidance, personalized adaptation, and trust-building feedback. We operationalize these principles through an AI-driven audio guidance system featuring: (1) standardized motion descriptions, (2) personalized pacing control, and (3) real-time corrective feedback. Our work provides insights for designing inclusive fitness experiences for BLV users.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; *Empirical studies in accessibility*.

Keywords

Workout Video, Body Exercise, Audio Description, Blind, Low Vision, Visual Impairment

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1 Introduction

The global online fitness market has experienced exponential growth, reaching \$12.18 billion in 2023 [8]. A key driver of this expansion is the widespread availability of online workout videos that have transformed how people exercise. Notably, over 30% of individuals with disabilities have attempted to use these videos for their daily workouts [12]. However, blind and low vision (BLV) users face significant accessibility barriers due to the visual-centric design of most fitness content. Current fitness videos predominantly rely on visual demonstrations [12, 18] while providing inadequate audio descriptions and accessible support [5, 38]. This reliance on visual information excludes BLV users from fully participating in digital fitness programs and obtaining their potential health benefits.

To understand the challenges BLV users face when exercising with workout videos, we conducted video co-watching sessions and interviews with eight BLV participants with varying levels of exercise experience. Participants engaged with fitness videos differing in both accessibility level and workout intensity. Our study identified **three key challenges** faced by BLV users during video-based exercise: (1) unclear motion instructions, (2) mismatched pacing, and (3) lack of performance feedback. To address these challenges, we propose **five design principles** to make workout videos accessible: clear movement descriptions, step-by-step breakdowns, rhythm guidance, personalized adaptation, and trust-building feedback. We operationalize these principles through an **AI-driven audio guidance system** that provides: (1) standardized motion descriptions, (2) personalized pacing control, and (3) real-time corrective feedback. The system translates visual action sequences into structured verbal instructions and offers timely guidance synchronized with the workout pace. Additionally, it delivers real-time performance feedback based on users' movements, enabling BLV participants to engage more effectively with online workout videos.

Our preliminary findings offer valuable insights for designing more inclusive and accessible fitness experiences for BLV users.

2 Related Work

Physical activity is vital for BLV users, as insufficient exercise significantly lowers their quality of life compared to sighted individuals [7]. Yet, they face greater challenges in maintaining regular workouts [28]. Prior research explores three main approaches: (1) gamified experiences to enhance engagement [1, 9, 22, 23, 27, 30, 32], though often relying on predefined, slow-paced routines with limited adaptability; (2) wearable devices, depth cameras, or sensors for motion guidance [11, 25, 26, 31]; and (3) virtual environments simulating real-world sports [10, 14]. These solutions often require extra hardware and space, creating cost and convenience barriers. To overcome this, we aim to develop a more diverse, lightweight, and low-cost solution to expand workout options and improve accessibility.

Compared to hardware-based solutions [3, 19, 21, 36, 37], follow-along workout videos offer greater convenience and variety, making them the dominant mode for online fitness [2, 17]. Although audio descriptions (AD) help BLV users understand video content, most are designed for film and general media [2, 16, 24, 33–35], with limited fitness-specific adaptation. In complex workouts [15], BLV users struggle to interpret movements [6, 13, 29], especially without interactive or real-time feedback. Moreover, how different workout types pose distinct accessibility challenges remains under-explored. Our work addresses these gaps by investigating barriers BLV individuals face when using follow-along workout videos.

3 Study Methods

To investigate the challenges of BLV users when exercising with workout videos, we conducted video co-watching exercises and follow-up interviews with eight BLV participants of varying exercise experience levels. Our study revealed three key challenges faced by BLV users during video-based body exercise.

Participants. We recruited eight BLV users (P1–P8; four male, four female) from an online support community. Their ages ranged from 22 to 81 ($M = 38.6$, $SD = 20.2$). Five participants exercised more than 90 minutes per week, while three less than 90 minutes per week. All participants were native Mandarin speakers. This study was approved by our institutions' IRB. Table 1 lists their demographics.

Workout Videos. We selected three workout videos representing varying accessibility levels [20] and exercise intensities. All videos were in Mandarin. The videos included: (1) **High accessibility / Low intensity**: a glute stretch routine¹ featuring detailed verbal instructions and slow movements; (2) **Moderate accessibility / Medium intensity**: a Baduanjin (八段锦) practice² with periodic verbal guidance and medium-paced movements; and (3) **Low accessibility / High intensity**: an integrated warm-up and HIIT session³ (i.e., two separate sections within the same video) featuring minimal audio cues and fast-paced exercises. Figure 2 presents screenshots alongside participants' performance.

¹<https://www.bilibili.com/video/BV1w4411Y7Kq>

²<https://www.youtube.com/watch?v=45vOPNkAcQU>

³<https://www.bilibili.com/video/BV1jk4y117eK>

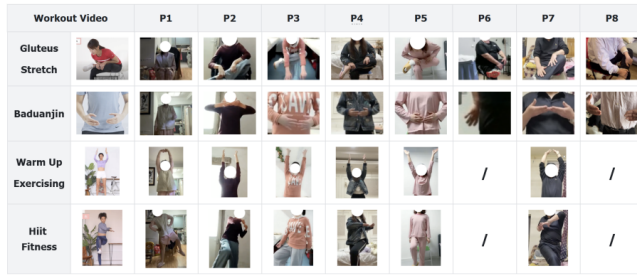


Figure 2: Video screenshots alongside participants’ performance for the three workout videos. Two participants did not complete the warm-up and HIIT session due to physical limitations (P6) and age-related constraints (P8).

Procedure. The study consisted of two successive phases: a 30-minute co-watching exercise followed by a 30-minute semi-structured interview, separated by a 5-minute break. During the co-watching exercise, We took notes on the challenges participants faced with the three selected workout videos. In the subsequent interview, the participants were asked to reflect on the challenges they encountered during video-based exercises. The study was conducted one-on-one via Zoom, and participants received assistance from sighted helpers to position their phones to ensure their body postures were visible throughout the study.

Analysis. We recorded the study videos, transcribed the audio, and applied an open-coding approach [4] to analyze the primary challenges BLV users face when engaging in video-based body exercise. Four researchers independently analyzed the transcripts, iteratively generating and refining codes until reaching an agreement. The resulting codes were then grouped into clusters that captured the key themes emerging from the data. Following a discussion of all possible themes, the final set of themes was identified and presented as the study findings.

4 Findings: Three Key Challenges

Our study identified three primary challenges BLV users face when engaging with online fitness videos.

Challenge 1: Unclear Motion Instructions. The lack of precise and structured descriptions prevents BLV users from fully understanding the instructional audio. During the study, seven participants reported difficulty understanding instructions, particularly those involving: (1) quantitative metrics (e.g., “raise your arm 30 degrees”), (2) metaphorical language (e.g., “move like a windmill”), and (3) technical terminology (e.g., “engage your core”). Additionally, videos that relied solely on on-screen texts were inaccessible to all participants. Individual differences in cognitive ability and prior exercise experience also affected comprehension—for instance, P3, who had extensive exercise experience, found the instructional videos easier to follow, whereas P2 struggled due to limited prior experience.

Challenge 2: Misaligned Exercise Pace. The absence of rhythmic cues and reflection time in the audio instructions makes it difficult for BLV users to keep pace with audio-based exercises. All participants reported struggling to follow the movements when: (1) the beat of each movement was not clearly marked, (2)

there was no indication of movement completion or transition, and (3) the fast pace and lack of pauses left no time for reaction.

Challenge 3: Lack of Performance Feedback. The lack of performance feedback prevents BLV users from practicing correctly and confidently. Participants highlighted difficulties due to: (1) insufficient breakdowns of continuous movements, (2) unclear guidance on coordinating body parts, and (3) a lack of detail on key posture elements like balance and core activation. Additionally, given variations in age and physical ability, users stressed the need for more flexible, fault-tolerant movement standards. Beyond these gaps, four participants expressed skepticism toward AI-generated feedback, questioning its accuracy and real-time responsiveness.

5 Design Principles

To address these challenges, we propose five design principles to support self-guided and efficient video-based exercise for BLV users:

(D1) Clear Movement Descriptions: Use concise language with body references (e.g., “lift hands to shoulder height”) and follow a consistent phrasing structure: name → beat → movement → body → direction → position.

(D2) Step-by-step Breakdown: Deconstruct complex movements into steps, allow time for response, and incorporate relational cues (e.g., “raise arms while bending knees”) to clarify coordination.

(D3) Rhythm Guidance: Align rhythmic cues (e.g., beats) with each movement transition to support timing and flow (e.g., [first beat] action 1, [second beat] action 2...).

(D4) Personalized Adaptation: Tailor instructions to users’ cognitive and physical abilities. The system should provide customization support for speech rate, feedback frequency, and error correction sensitivity.

(D5) Trust-Building Feedback: When providing feedback, the system should convey its confidence level and offer timely responses to help users assess accuracy and foster trust in its guidance.

6 AI-Driven Audio-Guided Workout Approach

Building on our findings, we found AI-driven solutions helpful for BLV users to better follow online fitness videos. As illustrated in Figure 3, the system integrates three core features:

Feature 1: Standardized Motion Descriptions. This feature enhances users’ understanding of movement instructions. The AI decomposes the exercise video, prioritizing the body part with the largest range of motion. It describes this primary motion using a structured format, then sequentially explains the movements of other body parts. Finally, it integrates these descriptions to clarify their coordination, enabling users to grasp the full-body movements.

Feature 2: Personalized Pacing Control. This feature supports users in maintaining rhythm and pace during workouts. The AI agent identifies natural pause points between movements and inserts adaptive pauses to provide sufficient time for comprehension and reaction. Users can also control playback speed or pause the session as needed, allowing them to proceed at a pace that suits their individual understanding and comfort.

Feature 3: Real-Time Performance Feedback. The system captures real-time motion data via the user’s device camera and compares it against the target movements in the original fitness

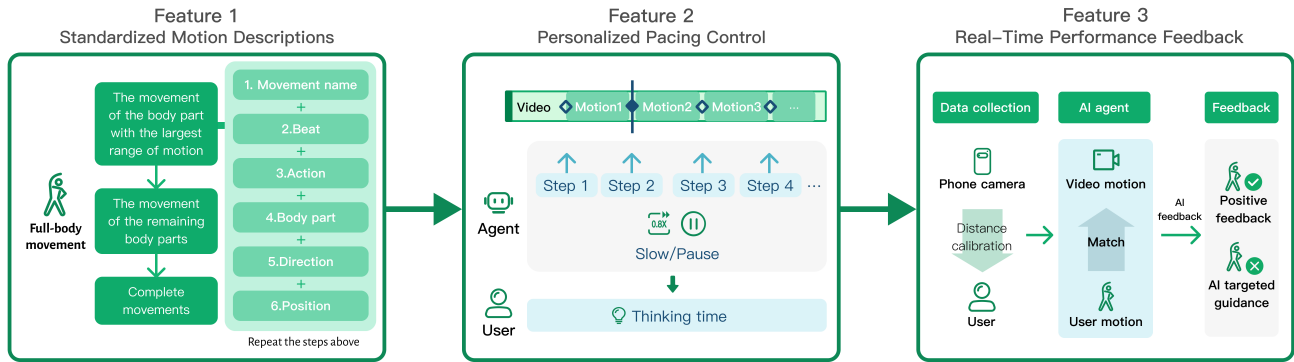


Figure 3: We propose an AI-driven, audio-guided approach to make video-based exercise more accessible for BLV users. It integrates three core features: (1) standardized motion descriptions, (2) personalized pacing control, and (3) real-time performance feedback.

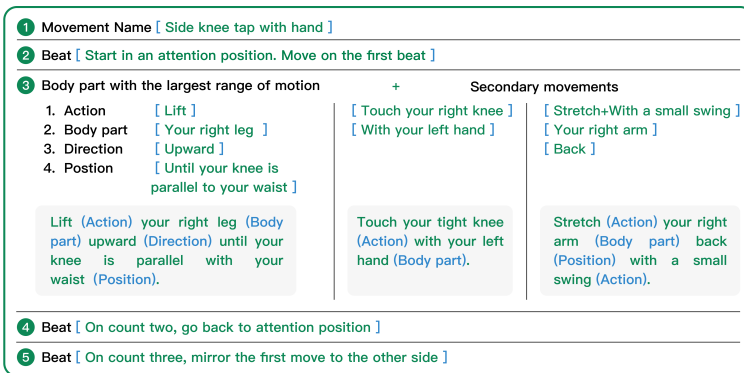


Figure 4: Example walkthrough of the *side-knee touch* exercise. The system delivers standardized motion descriptions, a step-by-step breakdown, and real-time feedback to support accessible video-based workouts.

video. Using a natural language model, the AI provides actionable verbal feedback. When movements are performed correctly, it delivers positive reinforcement. If more than four consecutive errors occur, the system prompts the user with the option to skip the current exercise, helping maintain flow and prevent discouragement.

Figure 4 presents an example walkthrough. The system first introduces the action name, *side-knee touch*, to provide users with a general understanding of the movement. It then provides a step-by-step movement breakdown aligned with the workout rhythm. On the first beat, the AI begins with the most prominent motion—the leg movement—to establish the core action for BLV users: “*Lift your right leg upward until your knee is parallel with your waist.*” Next, instructions for other body parts are provided: “*Left hand: Touch your right knee with your left hand. Right hand: Stretch your right arm back with a small swing.*” Finally, the system explains how these movements coordinate to form a cohesive body gesture. On the second beat, users are guided to return to the starting position. On the third beat, they mirror the movement on the opposite side, completing a full, symmetrical sequence.

7 Conclusion and Future Work

This study identifies three main accessibility challenges BLV users face with online workout videos: (1) unclear instructions, (2) mismatched pacing, and (3) lack of feedback. In response, we propose five design principles—clarity, breakdown, rhythm, personalization, and trust—to guide accessible exercise design. We operationalize these principles in an AI-driven audio-guided system that incorporates three core features: (1) standardized motion descriptions, (2) personalized pacing control, and (3) real-time corrective feedback.

In future work, we plan to conduct a comprehensive evaluation of the proposed system. Further works could also explore: (1) developing personalized workout support tailored to different age groups or experience levels, (2) implementing safety mechanisms to prevent injury, and (3) extending the system to support a wider range of workout types. Additionally, while the current design focuses on first-time video interactions, future systems could adapt the instructions based on users’ exercise history—for instance, offering more concise guidance for repeated workouts. Finally, future work should also consider potential privacy concerns. For example, systems could avoid storing workout recordings and ensure bystanders are not unintentionally captured, through local processing or privacy-aware design.

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A Supplementary Materials

This section includes the following tables and figures:

- Table 1 summarizes the demographic information of the participants.
- Figure 5 presents an illustration of the five design principles.

Table 1: Demographics of the participants. Visual onset refers to the time at which vision loss began, and is categorized as either congenital (from birth) or acquired (after birth)

ID	Age	Gender	Visual Condition	Visual Onset	Exercise Time Per Week
P1	22	F	Legally blind	Acquired	> 90 minutes
P2	23	M	Totally blind	Congenital	< 90 minutes
P3	25	F	Totally blind	Congenital	> 90 minutes
P4	25	F	Legally blind	Acquired	> 90 minutes
P5	29	F	Legally blind	Congenital	< 90 minutes
P6	44	M	Totally blind	Acquired	> 90 minutes
P7	60	M	Legally blind	Acquired	< 90 minutes
P8	81	M	Totally blind	Acquired	> 90 minutes

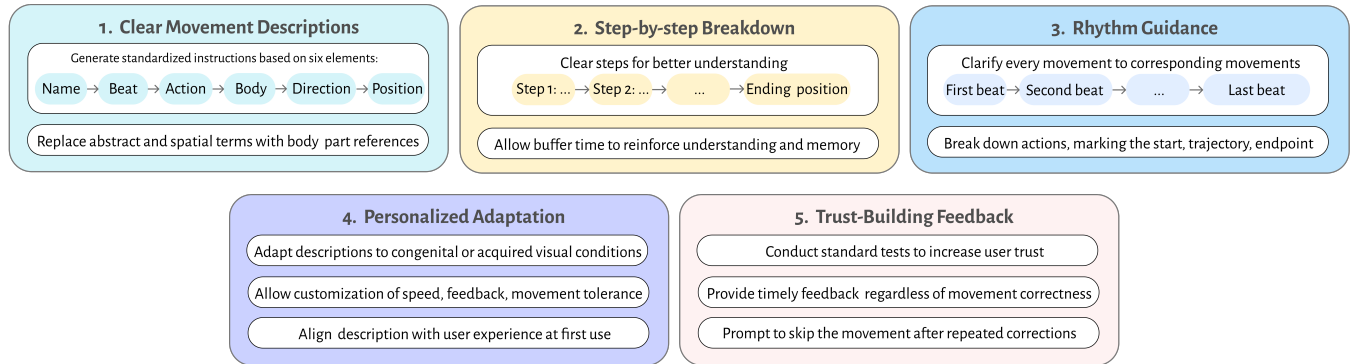


Figure 5: Five design principles to support independent and effective video-based exercise for BLV users.